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Studierichting: Informatica
Oefeningenreeks 6D nr. 31.3

$$\sum_{n=2}^{\infty} \frac{\ln n}{n}$$

1. Op $[2, +\infty[$: $f(x) > 0$

2. Dalend: $f'(x) = \dfrac{1 - \ln x}{x^2} < 0$

$\Rightarrow$ integraalcriterium

$$\int_2^{+\infty} \frac{\ln x}{x} dx = \lim_{b \to +\infty} \int_2^b \frac{\ln x}{x} dx = \lim_{b \to +\infty} \left[\frac{(\ln x)^2}{2} \right]_2^{+\infty} = +\infty$$

Dus $\displaystyle\sum_{n=2}^{\infty} \frac{\ln n}{n}$ is divergent

References